

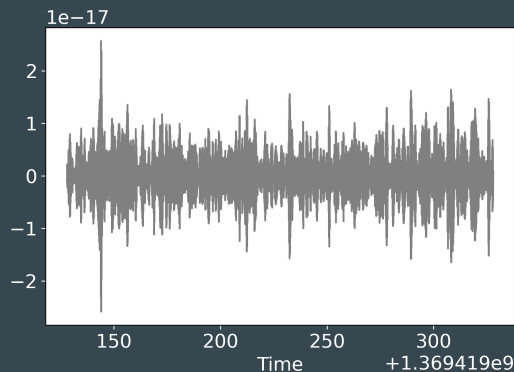
How to know I can(not)
trust my GW inference?

...

(Usually it is better not to)

Parameter estimation for gravitational waves

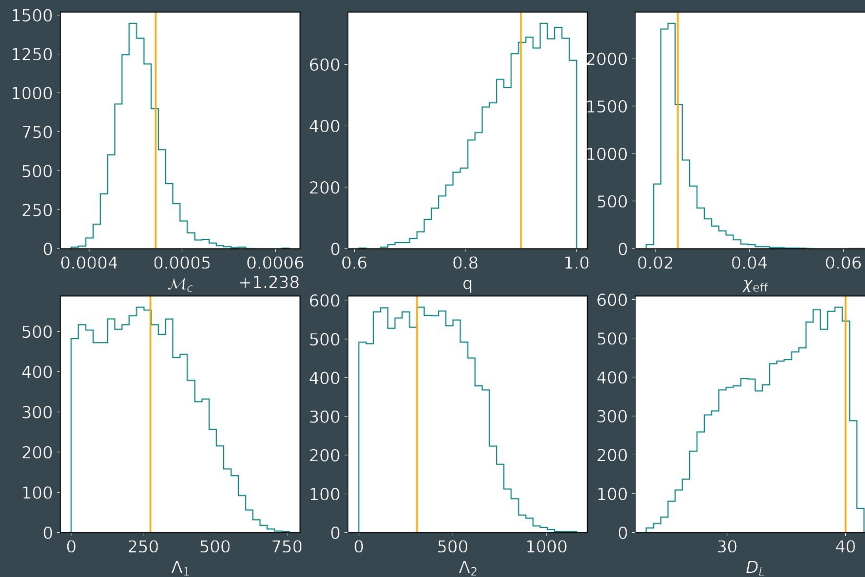
Source parameters $\longleftrightarrow$ GW signal



Parameter
estimation

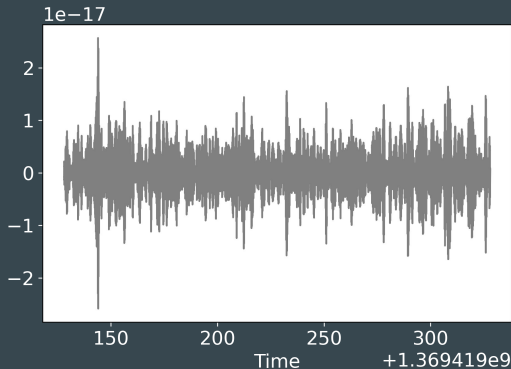


Bayesian inference



Parameter estimation for gravitational waves

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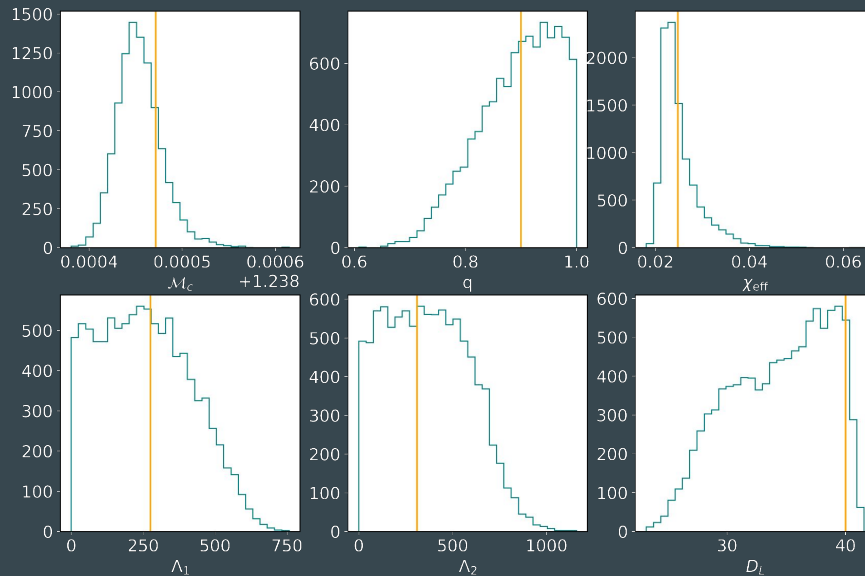


Bayesian inference



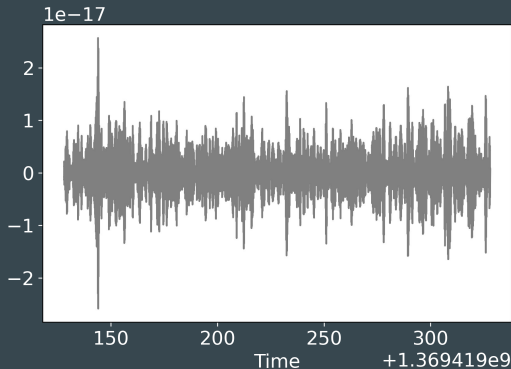
Bayes theorem:

$$p(\vec{\theta}|\mathcal{H}_s, d) = \frac{p(d|\vec{\theta}, \mathcal{H}_s)p(\vec{\theta}|\mathcal{H}_s)}{p(d|\mathcal{H}_s)}$$

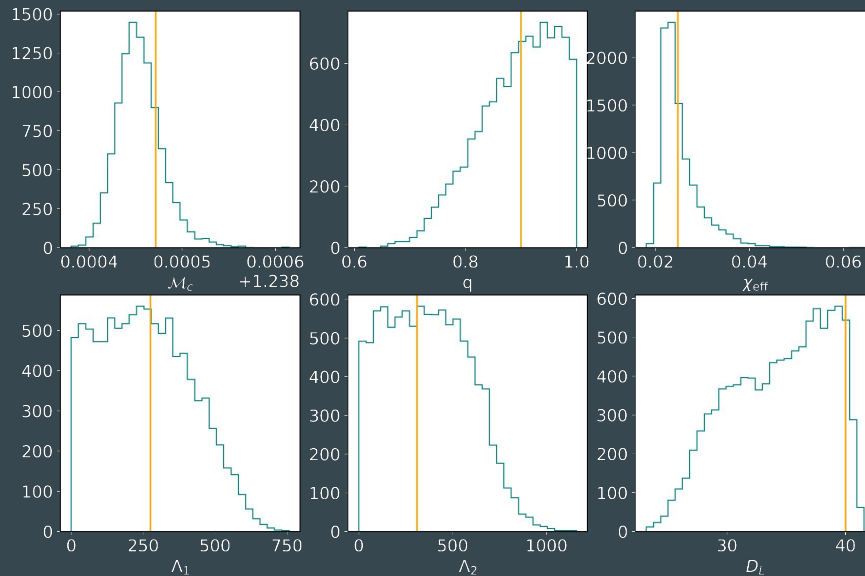


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Bayesian inference ✨

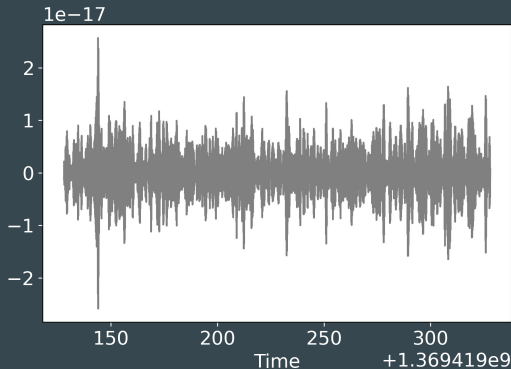
Bayes theorem:

posterior

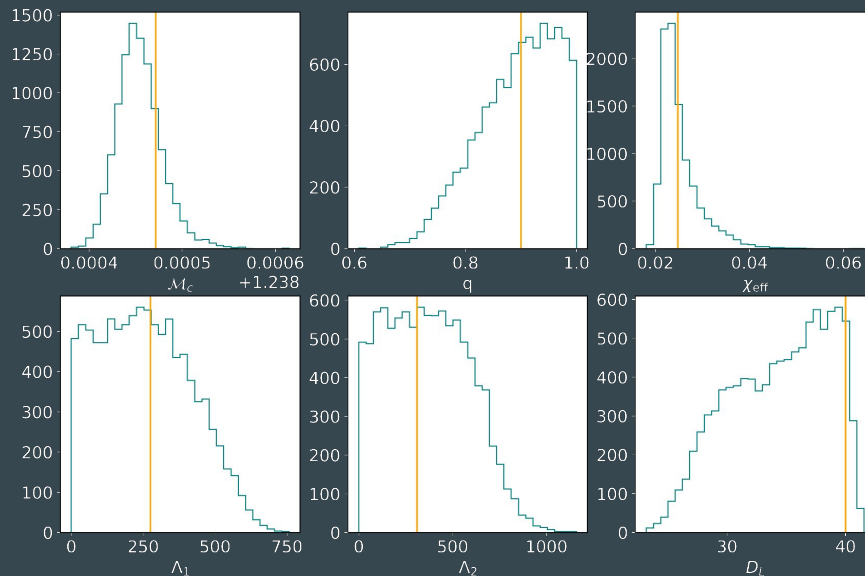
$$p(\vec{\theta}|\mathcal{H}_s, d) = \frac{p(d|\vec{\theta}, \mathcal{H}_s)p(\vec{\theta}|\mathcal{H}_s)}{p(d|\mathcal{H}_s)}$$

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Parameter
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Bayesian inference ✨

Bayes theorem:

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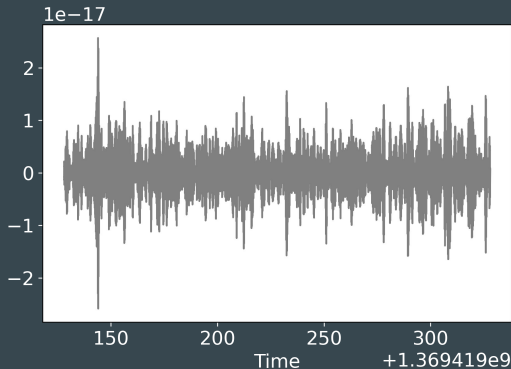
$$p(\vec{\theta} | \mathcal{H}_s, d)$$

likelihood

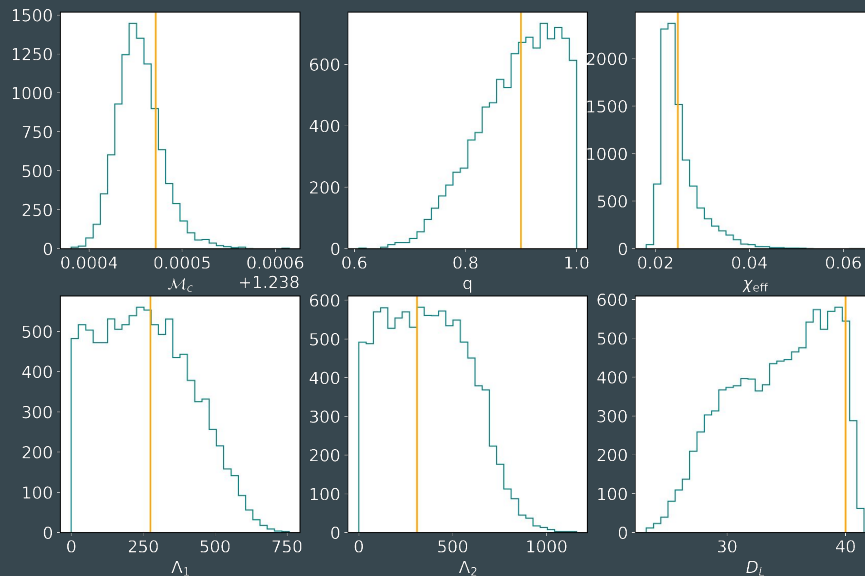
$$= \frac{p(d | \vec{\theta}, \mathcal{H}_s) p(\vec{\theta} | \mathcal{H}_s)}{p(d | \mathcal{H}_s)}$$

Parameter estimation for gravitational waves

Source parameters $\longleftrightarrow$ GW signal



Parameter
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Bayesian inference ✨

Bayes theorem:

posterior

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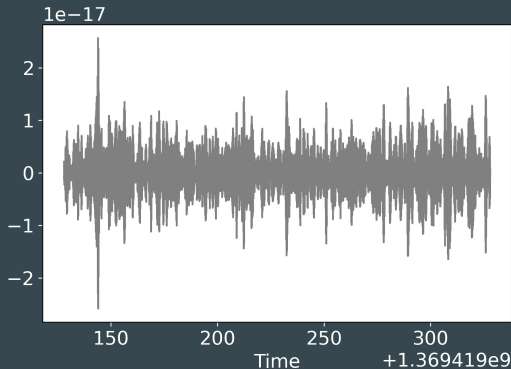
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likelihood

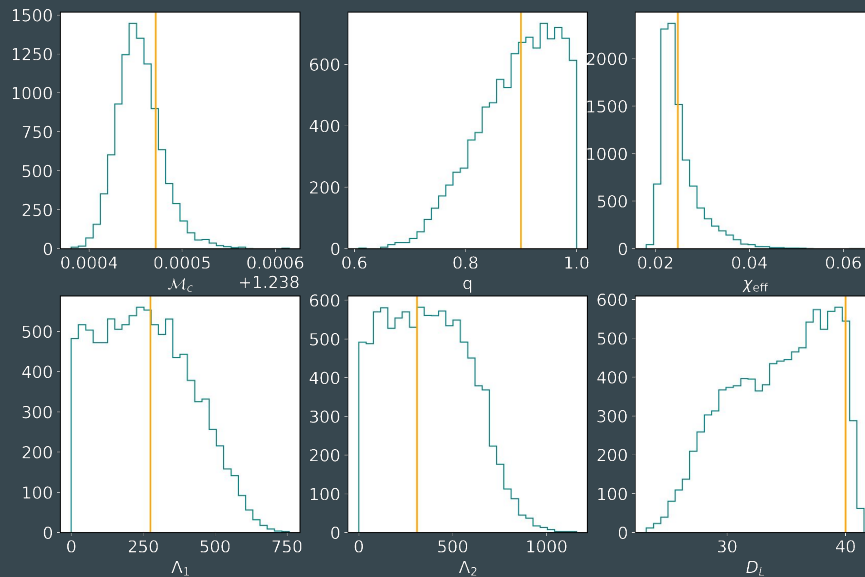
prior

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Parameter
estimation



Bayesian inference ✨

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likelihood $\swarrow$ prior $\nearrow$

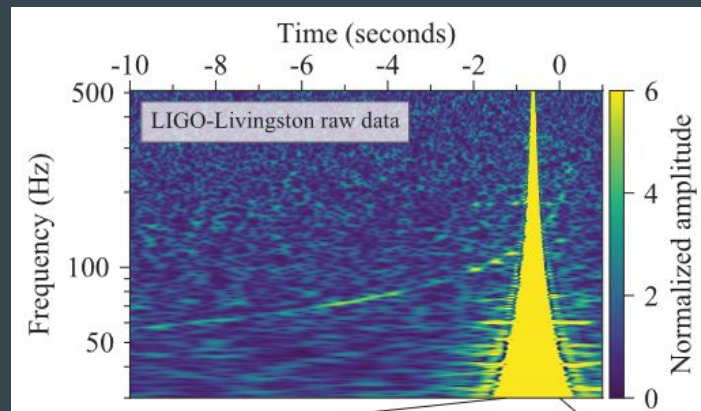
posterior $\swarrow$ evidence $\searrow$

Potential issues in GW inference

$$p(\vec{\theta}|\mathcal{H}_s, d) = \frac{p(d|\vec{\theta}, \mathcal{H}_s)p(\vec{\theta}|\mathcal{H}_s)}{p(d|\mathcal{H}_s)}$$

Potential issues in GW inference

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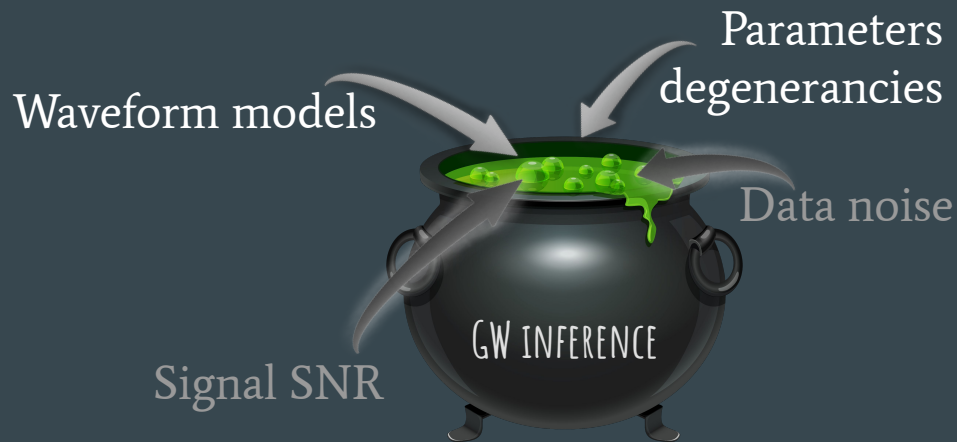
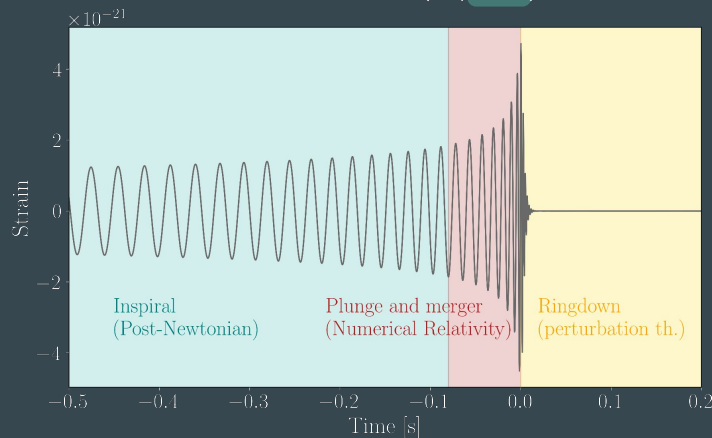


louder signals => better constraints (but also systematics)

Potential issues in GW inference

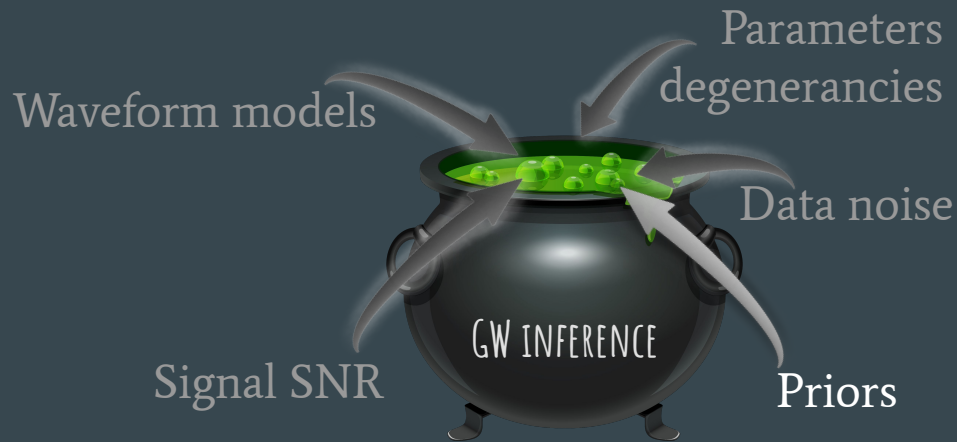
- different families (NRsurrogates, Phenom, EOB)
- different content (precession, HM, eccentricity, matter effects...)

$$p(\vec{\theta} | \mathcal{H}_s, d) = \frac{p(d | \vec{\theta}, \mathcal{H}_s) p(\vec{\theta} | \mathcal{H}_s)}{p(d | \mathcal{H}_s)}$$



Potential issues in GW inference

$$p(\vec{\theta}|\mathcal{H}_s, d) = \frac{p(d|\vec{\theta}, \mathcal{H}_s) \underbrace{p(\vec{\theta}|\mathcal{H}_s)}_{\text{Priors}}}{p(d|\mathcal{H}_s)}$$

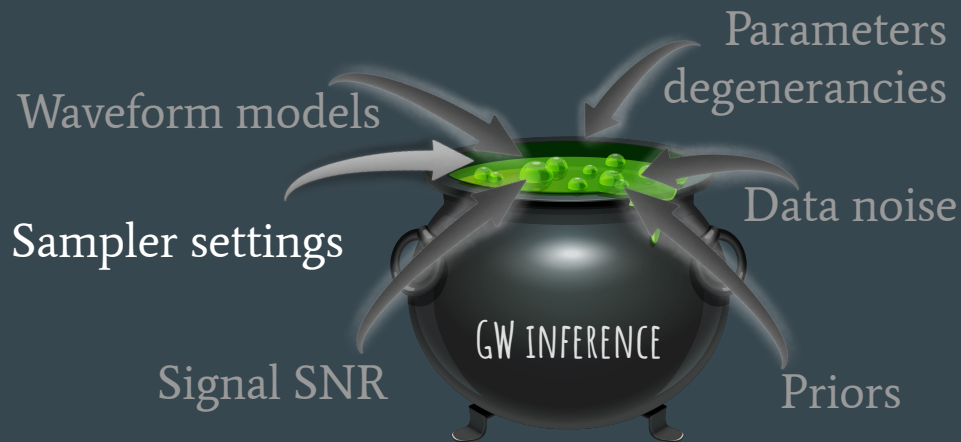


Potential issues in GW inference

Not feasible analytically

=> stochastic **samplers** (nested, MCMC)

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Potential issues in GW inference

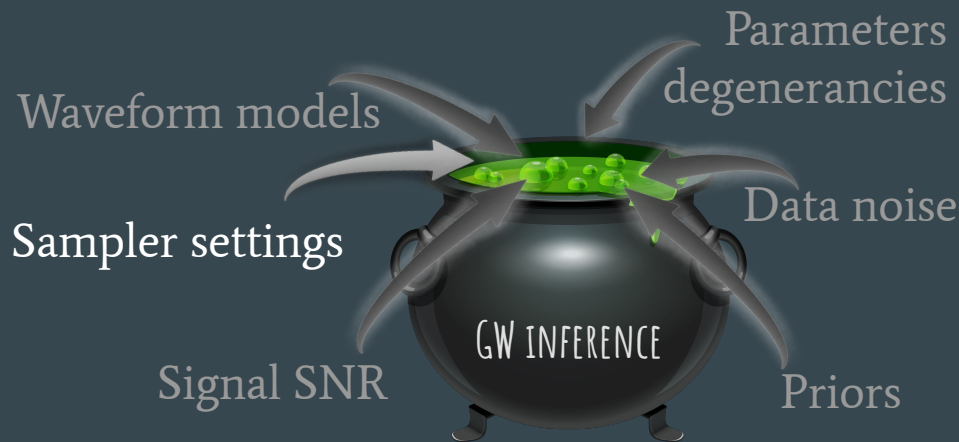
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Sampler:

- which one?
- settings? (nlive, temperatures, chains, convergence...)
- might get stuck or struggle to sample 'difficult' distributions (e.g., multimodal)



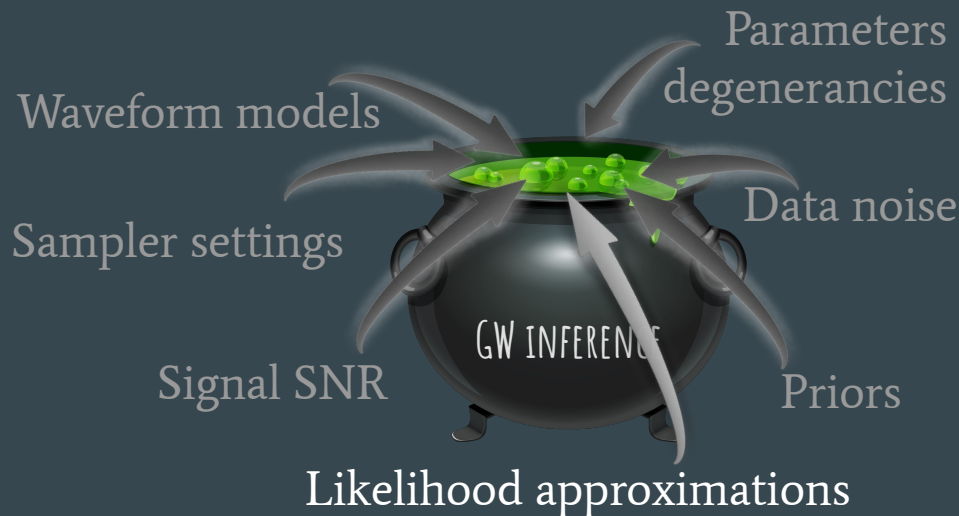
Potential issues in GW inference

$$p(d|\vec{\theta}, \mathcal{H}_s) \propto \exp\left[-\frac{1}{2}\left\langle d - h(\vec{\theta}) | d - h(\vec{\theta}) \right\rangle\right]$$

$$\langle a|b \rangle \equiv 4\Re \int_{f_{\text{low}}}^{f_{\text{high}}} \frac{\tilde{a}^*(f)\tilde{b}(f)}{S_n(f)} df$$

$$p(\vec{\theta}|\mathcal{H}_s, d) = \frac{p(d|\vec{\theta}, \mathcal{H}_s)p(\vec{\theta}|\mathcal{H}_s)}{p(d|\mathcal{H}_s)}$$

Computationally very expensive
=> **approximation** methods



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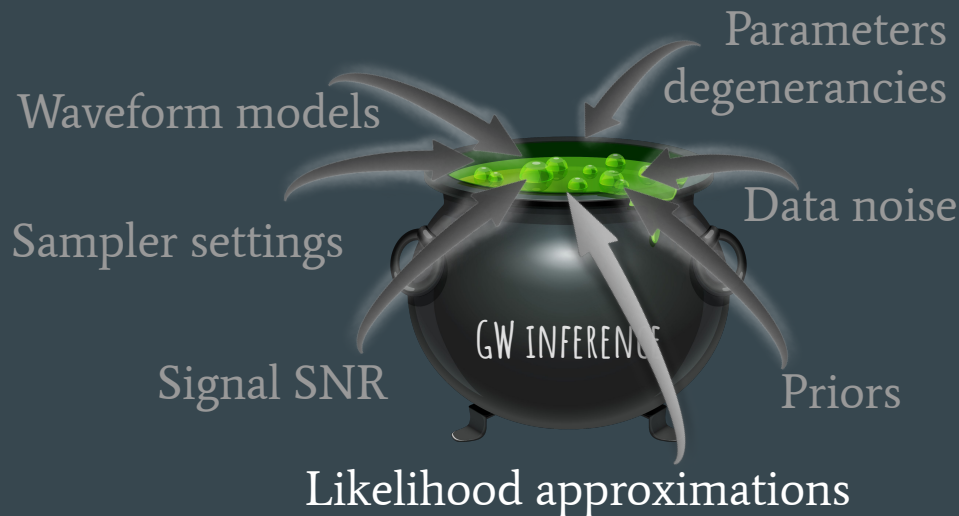
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Likelihood approximations:

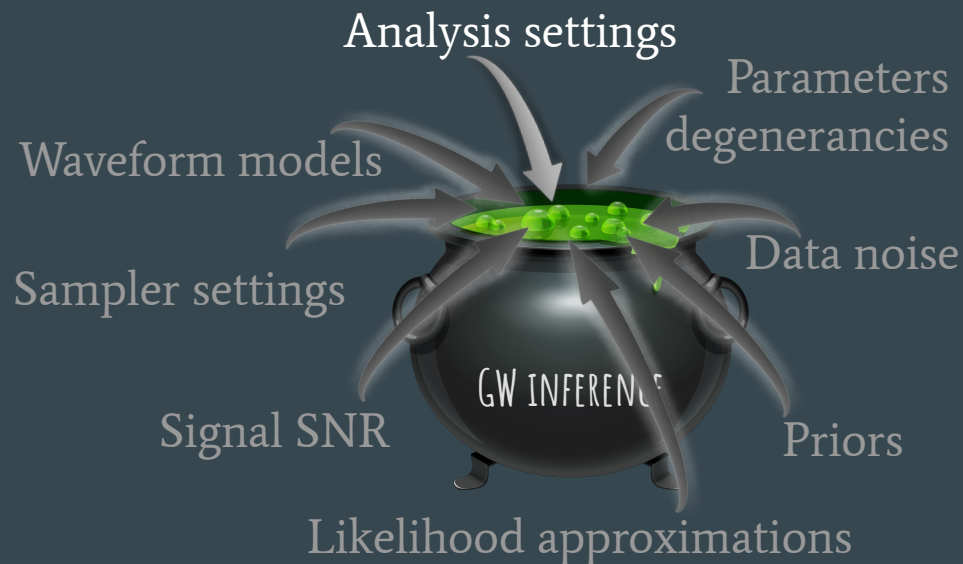
- multibanding: frequency grid based on time to merger
- relative binning: evaluate the ratio to a 'fiducial' waveform on a given grid
- ROQs: basis-decomposition of likelihood over the parameter space pre-computed, evaluate sample likelihood via interpolation

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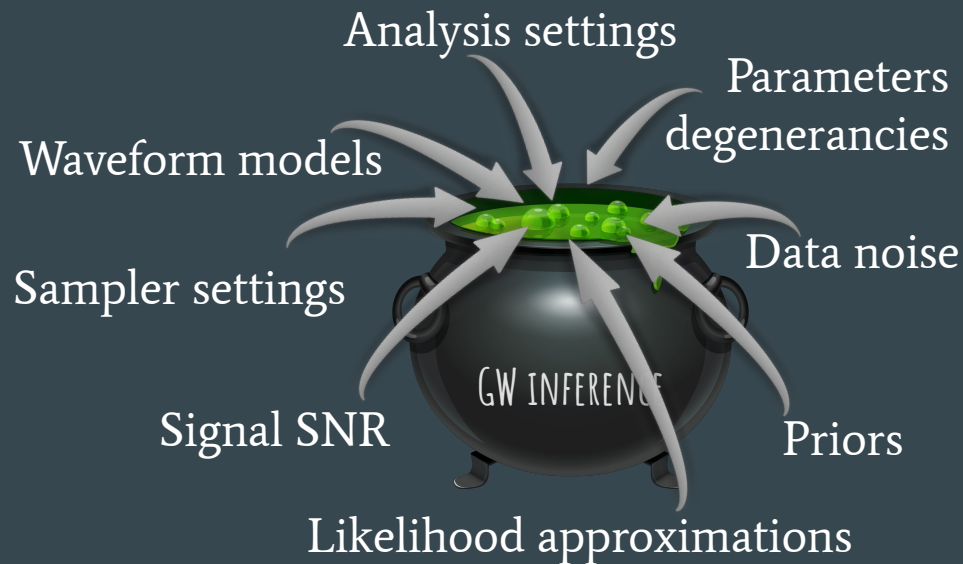
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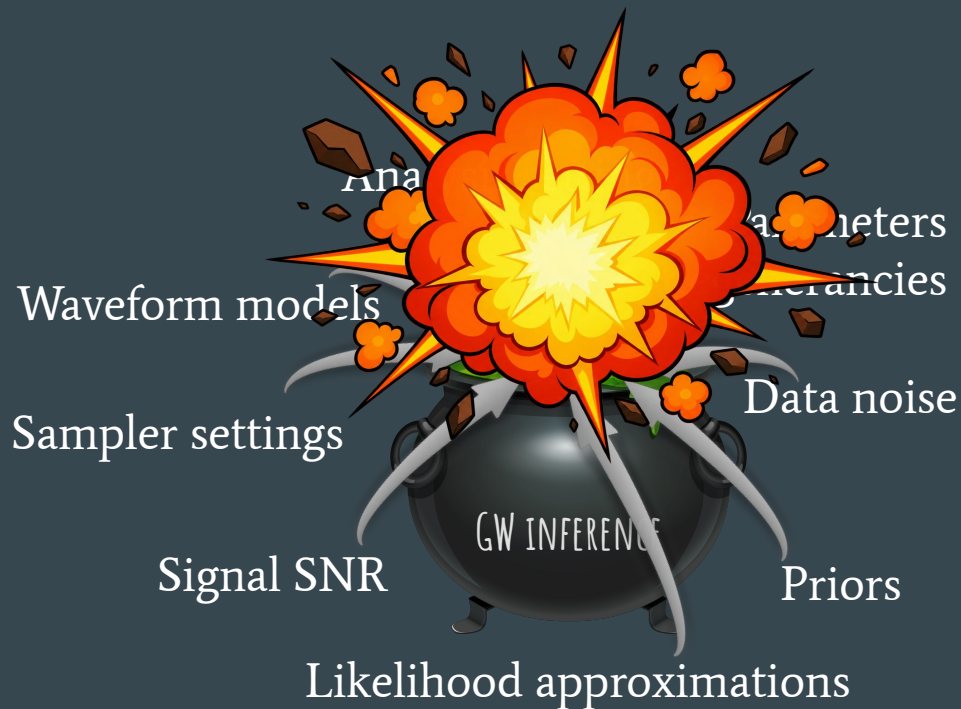
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Go to www.menti.com
Enter the code
6799 1280

“What’s wrong with you?” - GW posteriors edition

